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Description - Contains theory for the 1.0T Body Coil. Sheet numbers listed at the end of some subsection titles refer to the sheet number within the TPS/RF (1.0T) Block Diagrams.

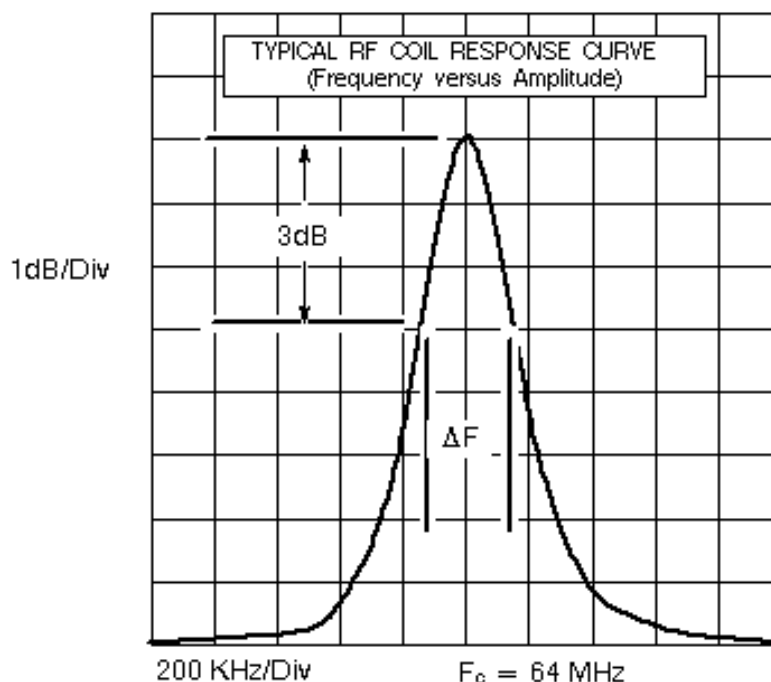
1- INTRODUCTION (BODY COIL)

The function of the radio frequency (RF) coils used in Signa MR systems is to either transmit RF energy into the nuclei (of atoms) of a patient, receive the small MR signal from the patient, or both. This received signal (raw data) will later be reconstructed into an image. There are different RF coils depending on the application being performed. A large body coil is capable of whole body imaging, while smaller coils, such as the one used for heads are designed specifically for a smaller area of interest.

Some RF coils can be used to both transmit and receive as with the body, head, and extremity coils. Surface coils are designed to only receive. Basic RF coil design concepts include quality factor (Q), loading effects, and decoupling.

1-1 Quality Factor (Body Coil)

The quality factor (Q) is typically a measure of the bandwidth of an RF coil. The Q is calculated by dividing the center frequency (F_c) of the RF coil response, by the frequency difference (ΔF =bandwidth) between the half power points (-3dB from peak power) on a response curve. The MR RF body coils are designed for an unloaded (empty) Q of approximately 200. This means that the ΔF is approximately 320KHz (with a center of 64 MHz). See Illustration 1-1. The higher the Q, the narrower the bandwidth. The higher the Q, the more selective the coil is. This selectivity is actually a band-pass filter that reduces the amount of noise and spurious signals that are received by the RF coil and passed on to the receiver hardware.

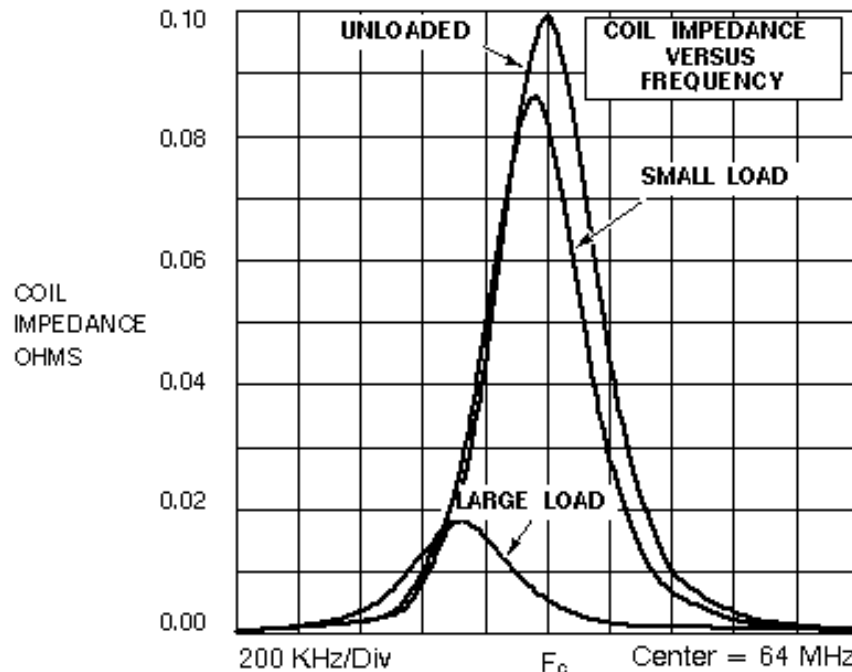


$$Q = \frac{(F_c)}{(\Delta F)}$$

RF COIL QUALITY FACTOR
ILLUSTRATION 1-1

1-2 Loading Effects (Body Coil)

The "Q" is affected by the load on the RF coil. A load can be an image quality phantom, body, head, or an extremity. Different loads with different compositions (water versus muscle versus fat) have different loading characteristics. Larger loads of the same composition have more of an effect than smaller ones. The placement of a load in respect to the RF coil also has a great effect on the Q. When an RF coil becomes loaded, the RF coil response broadens and the Q decreases as shown in Illustration 1-2. The decreased Q results in less selectivity by the coil which in turn reduces the level of the received signal.



RF COIL LOADING EFFECTS
ILLUSTRATION 1-2

There is also a frequency shift that is associated with the loaded response. However, with large loads because the Q has decreased (and the bandwidth has increase by approximately six times), the RF coil is still for all practical purposes operating at an acceptable frequency. If the RF coil were to be tuned for a loaded condition, then a smaller load which does not shift the RF coil frequency as much and does not broaden the bandwidth would not be operating at an acceptable frequency. This would result in poor image quality. This is why RF coils are designed to be tuned unloaded. It produces the best compromise for all types of loads.

The load on a coil is very sensitive to the position of the patient with respect to the RF coil. The body acts like a capacitor. This capacitance can add to that of the coil and shift the frequency down. In a quadrature type coil, this capacitance can affect one mode (vertical or horizontal) of the coil to the point where it is no longer operating in quadrature (a rotating RF field). The coil is then operating in a linear mode which requires twice as much power to prescan and also has 40% less signal to noise. This effect is minimized by using small pads between the patient and the coil. This pad is actually a small capacitor in series with the body's large capacitance, which effectively cancels the body's effect.

If a RF coil is only being used to transmit and the receive function is being performed by a separate coil, then there must be a mechanism to decouple the transmit coil from the receive coil during the receive time. Without decoupling, the receive area is increased and the benefits of a having a smaller, localized coil are compromised. Not having proper decoupling could cause an increase in noise and a decrease in SNR.

1-3 Receive Coil Design (Body Coil)

The goal of a receive coil is to maximize the signal to noise ratio (SNR). Noise is a function of temperature, resistance, and system bandwidth. Because resistance increases with an increase in the coil receive area, an ideal coil is one that is only as large as is needed to scan a specific part of the anatomy. This is why receive only surface coils have better SNR performance. They are smaller and typically customized for a specific type of anatomy (i.e. neck, eye, and breast surface coils).

Quadrature design is also important for a receive coil. The quadrature effect will increase the level of signal while at the same time decrease the noise level. This is due to the signal being correlated (in phase) while the noise is uncorrelated (random - not in phase) in the rotating RF field. This improves the SNR for a quadrature type coil over a linear one by a factor of 1.4.

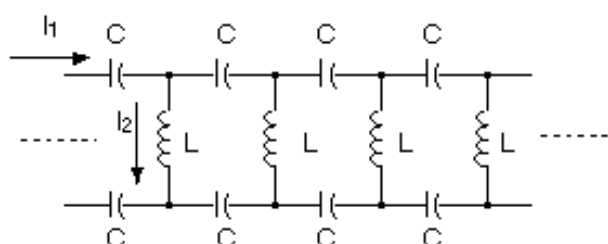
Receive only coils should be designed so that they are detuned during the transmit signal. This is accomplished with what is referred to as a blocking network. When the blocking network is working, the RF coils frequency response is altered so that it no longer resonates at the transmit frequency. If the blocking network is not functioning it is possible for several things to happen including: flip angles can not be properly calibrated, the surface preamplifier could be damaged due to high signal levels, and the patient could be exposed to high concentrations of RF on the surface coil.

2- RF COIL DESIGN (BODY COIL)

The discussion of the basic ladder design and RF fields can be applied to both the linear and quadrature type coils, whether they be body or head. All MR RF coils are based on these two basis building blocks. Specific hardware pertaining to the tuning and operation of the RF coils can be found in the TPS/RF tabs. Refer to them for additional information.

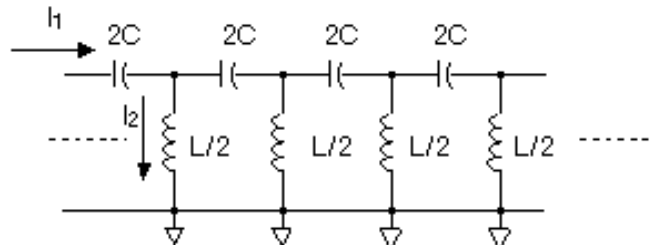
2-1 Ladder Networks (Body Coil)

The body and head Coils used for the Signa system are based on the ladder network design. A ladder network can be classified as either a high-pass or low-pass. A ladder network which has series capacitors and parallel (shunt) inductors is considered a high-pass. See Illustration 2-1. Conversely, a ladder that has series inductors and parallel (shunt) capacitors is a low-pass. A real ladder network will usually have both capacitive and inductive elements in all sections but, one of the two elements will dominate the other.



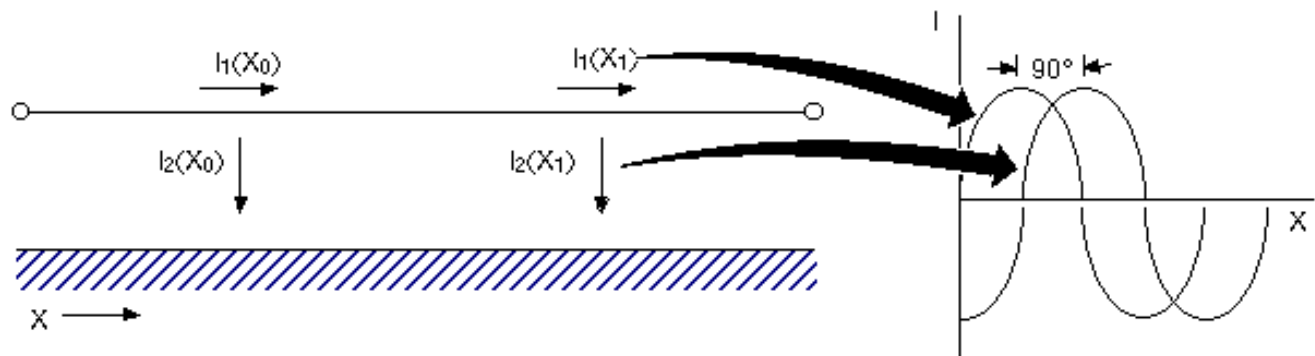
1
NORMAL HIGH PASS LADDER NETWORK
ILLUSTRATION 2-1

If a high-pass ladder network is folded onto itself the resulting elements are now twice the capacitance and one half the inductance. A ground plane will also result from the new configuration. See Illustration 2-2.



FOLDED HIGH PASS LADDER NETWORK
ILLUSTRATION 2-2

This new ladder has the characteristics of a lumped element transmission line. The series and shunt currents in this transmission line vary along the length. The function of this varying current is sinusoidal along the length (X) of the transmission line. The series and shunt currents are also out of phase with each other. Series currents which vary as $\sin(x)$ will have shunt current that vary as $\cos(x)$, or 90° out of phase with each other. See Illustration 2-3.



TRANSMISSION LINE EQUIVALENT
ILLUSTRATION 2-3

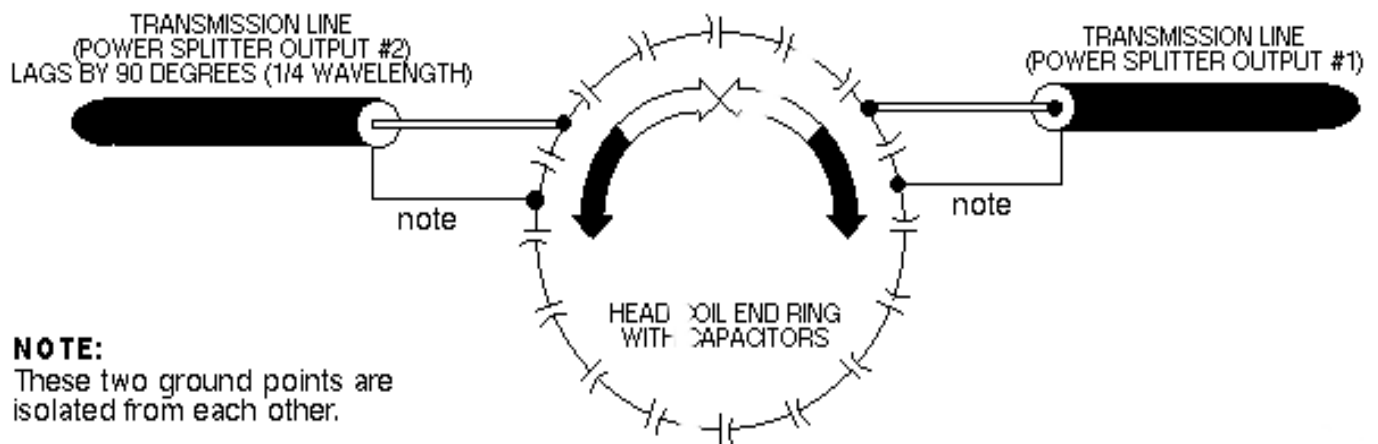
2-2 Bird Cage Coil (Body Coil)

The 1.0T Quad Body Coil can be thought of as a high-pass ladder network wrapped around a cylindrical form with its ends connected together. This results in a design that looks much like a bird-cage. The series capacitors form a ring at the ends of the coil and the parallel (shunt) inductors are horizontal runs. The actual physical elements in a Quad Head Coil are chip capacitors placed at the end rings and copper runs on a substrate material.

Because an RF Coil (Quad Body or Head) is a ladder network that has both its ends connected together, certain conditions must exist. One condition is that the current in one element has to equal that in the next. This means that a sinusoidal (or RF) wave has to be continuous. This sets up an environment where only certain frequencies of a sinusoidal (or RF) wave can exist (or resonant). In the Body Coil this resonant frequency(s) is a function of both the chip capacitors placed at the end rings and the length (or inductance) of the bird-cage.

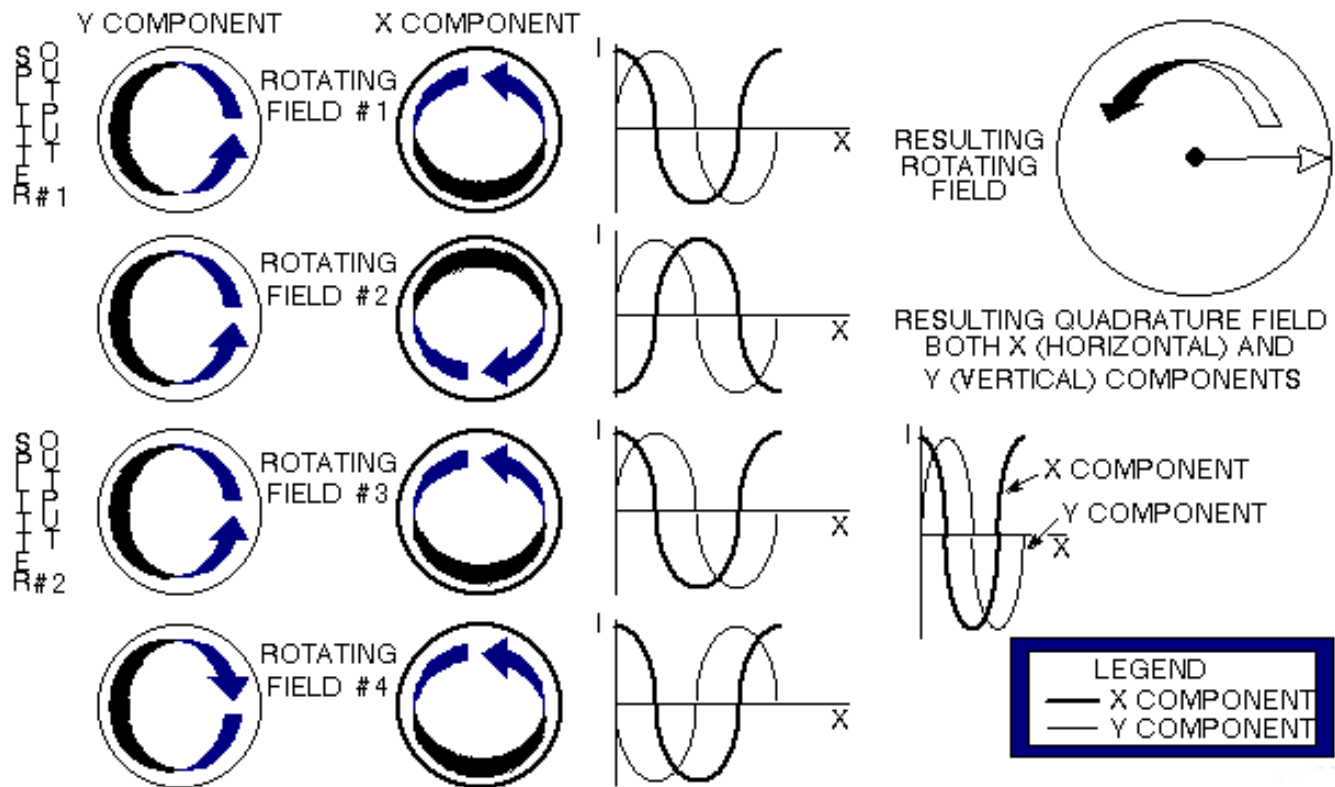
2-3 Quadrature Operation (Body Coil)

A quadrature designed coil uses two RF inputs, with one delayed by 90° (or $1/4$ wavelength), and connected to the coil 90° from each other to produce a rotating RF field. See Illustration 2-4.



QUADRATURE COIL RF INPUT SIGNALS
ILLUSTRATION 2-4

This design is based on the fact that destructive fields (ones that do not help the MR process) will cancel each other, leaving only the X and Y components that produce a rotating RF field inside of the coil. See Illustration 2-5.



QUADRATURE COIL RF FIELDS
ILLUSTRATION 2-5

3- BODY QUAD HYBRID ASSEMBLY - SHEET 9A

The 1.0T Body coil is a High-pass birdcage structure with direct drive. The Body coil employs a direct drive configuration which requires no drive disable. The coil is a fixed tune coil with the tuning capacitors soldered directly to the Body coil. The Dynamic Disable Switch Boards (8) are mounted in the interior of the Body coil behind panels.

The Body splitter box consists of an integral Quad Hybrid/TR network with the Body coil preamplifier mounted to it. The Hybrid portion consists of C89, C90, C95, C96, C97, C98 and the associated microstrip transmission lines. These components are configured as a modified lumped element hi-pass Quad Hybrid. An input signal applied to the upper left port (junction of C90 and C95) will be split into two signals of equal magnitude and 90 degree phase difference and output to the I (J3) and Q (J4) ports. During receive, the echo signal is applied to the I and Q ports. The phase of this receive signal is such that the I and Q ports are summed and appear as an output signal on the isolated port (junction of C95 and C97).

The TR Switch portion of the hybrid assembly consists of a transmit mode and a receive mode. In the transmit mode, a DC bias current of 2.5 - 3.0 amps is input through the RF amp port, J1, and biases all the pin diodes to the "ON" state. In the receive mode, -5 volts is applied to J1 and turns all the pin diodes OFF.

4- DIRECT DRIVE DISABLE ASSEMBLY - SHEET 9A

The Body coil used a direct drive circuit. The direct drive disable switches disconnect the Body coil from the Quadrature Hybrid and associated circuitry at all times during head scans.

During body scans the PIN diodes are biased, applying a short circuit to ground at the end of a quarter wave stub. This short is reflected as an open circuit and RF energy passes unimpeded between the coil and the hybrid. During head scans the PIN diodes are biased in the opposite direction. The quarter wave stub reflects the open as a short. The drive loop cannot absorb significant energy if it "sees" a high impedance at its input.

5- QUAD II/ROEMER BODY HYBRID/TR/IDD ASSEMBLY - SHEET 9B

The Body Hybrid, T/R Switch, and Inductive Drive Disable Switch Boards are combined into the Body Hybrid/TR Assembly.

The Quadrature Hybrid Board for the Body Coil splits the output of the RF amplifier into two signals that are ninety degrees out of phase with one another. These two signals (I and Q) are applied to the inputs of the Body Coil, where a rotating RF field is produced. Reflected power from the coil appears at the isolated port of the Hybrid and is dissipated in the 200 Watt load which is located on the assembly.

A Body TR Bias signal is applied to the heliax cable at the RF amplifier EFB Assembly. This forward biases PIN diodes during transmit to couple the RF amplifier output to the Hybrid input and to couple the reflected power signal to the internal load. Two PIN diodes are also used at the input of the Body Preamplifier. The PIN diodes provide a short circuit to ground at the end of two quarter wave line sections between the Preamplifier input and the Quadrature Hybrid protect the Preamplifier from damage during the transmit cycle.

During receive, all PIN diodes are reverse biased. The RF amplifier and load are therefore disconnected from the Hybrid so that all the received signal from the coil is applied to the input of the Preamplifier for best receive sensitivity. The two PIN diodes at the preamplifier input act as open that essentially provide a direct path for the received signal.

Each inductive drive loop inside the quadrature drive Body Coil is a series resonant circuit. If the loops are connected to a low impedance, they will absorb RF energy from adjacent resonant coils, such as head or surface coils. This causes unacceptable reduction in received signal strength and signal to noise ratio. Coupling to the head coil can also result in damage to the Body Preamplifier from transmitted energy during head scans.

The inductive drive disable switches disconnect the drive loops from the Quadrature Hybrid and associated circuitry at all times during head scans.

Each switch consists of a "tee" at the coil input connector with a quarter wave stub to ground via a PIN diode switch. During body scans the PIN diodes are forward biased (by the COIL SWITCH BIAS signal), applying a short circuit to ground at the end of the quarter wave stub. This short is reflected as an open circuit to the "tee" and RF energy passes unimpeded between the coil and the hybrid. During head scans the PIN diodes are reverse biased. The quarter wave stub reflects the open as a short at the "tee" (coil input). Another quarter wave stub inside the coil reflects the short at the coil input as an open at the inductive drive loop. The drive loop cannot absorb significant energy if it "sees" a high impedance at its input.

6- DYNAMIC DISABLE SWITCH BOARD - SHEET 9

The Dynamic Disable Switch Boards are mounted in the interior of the Body coil behind panels. There are eight boards altogether. Five are located in the bore above the cradle and three are located below the cradle.

When these switches are placed in the body mode (with +400VDC reversed bias) they force the response of the Body coil to resonate close to magnet frequency. The exact frequency has been fixed by the board circuitry and additional tuning is not necessary.

When in head mode, the switches are forward biased with -15VDC. This has the effect of detuning the Body coil. Therefore, when inside of the Head coil, the predominant RF field is that of the Head coil itself. The exact frequency has been fixed by the board circuitry and additional tuning is not necessary.

7- BODY PREAMPLIFIER - SHEET 9A, 9B

The echo from the patient is extremely small in amplitude. To prevent it from being obscured by noise, it is necessary to boost its level before transmitting it along the coaxial cable from the exam room to the receiver. This is the function of the preamplifier. Identical units are provided for each coil in the exam room. The head coil preamplifier is also used for surface coils. For convenience, the power for each preamplifier is provided via its output coaxial cable from the inputs of the TPS Receiver Module. Since only one input at a time will be selected in the receiver, only one preamplifier will receive power. Thus operating power and preamplifier selection are provided with no additional penetrations into the exam room. The gain of each preamplifier is ~33dB.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 28, 1998	M. Whitlow	Initial conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document, changed illustration numbers per style guide.